
Q/R/M/C: Stage Decomposition for Memory-Based Navigation in Single Agents and Cadence-Coupled Collectives

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Abstract

1 Success-only metrics collapse qualitatively different memory-based navigation fail-
2 ures into a single scalar, hiding *which stage* of the retrieval pipeline is responsible.
3 We propose the Q/R/M/C decomposition — **Q**uery formation, **R**etrieval satisfac-
4 tion, target **M**aterialization, **C**ompletion after retrieval — and show that, under a
5 conditional stage-Markov assumption, the four stage factors are estimable directly
6 from standard trajectory logs (Proposition 1). **Single-agent.** On a 10-seed Mini-
7 Grid benchmark the framework localizes bottlenecks: hazard recovery is retrieval-
8 limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$) while goal-region rejoin is
9 downstream-limited (oracle retrieval rescues semantic-path completion without
10 moving end-to-end success). Inspection of the framework’s own logs refines the at-
11 tribution further: candidate *ranking* is near-optimal while candidate *generation* fails
12 at 91% of decision points, exposing a memory-accumulation limit. **Multi-agent.**
13 On a 35,640-run sweep with broadcast cadence $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$
14 and 40 seeds, the bottleneck-shift signature appears as a strongly negative within-
15 configuration correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ (Pearson -0.61 at
16 $K = 4$, decaying to -0.02 at $K = 64$), and mean time-to-success has an interior
17 minimum at $K = 8$. **Honest limit.** The strict interior-maximum corollary on the
18 conditional product $P(M^*|R^*) \cdot P(C^*|M^*)$ does *not* hold — the product mono-
19 tonically approaches the independent-agent boundary. The slope-level claim of
20 Proposition 3 is supported while one of its corollaries is not, and we report the
21 gap explicitly. Across both regimes the diagnostics converge on a single binding
22 constraint — how trajectory evidence is consolidated into persistent, shareable
23 symbolic structure — motivating *collective semantic memory* as the engineering
24 target that the framework is built to measure.

1 Introduction

25
26 Classical navigation benchmarks often assume that the agent is given an explicit destination coordinate
27 or a dense task reward. Many practical navigation problems are different: the agent must reach a
28 place of a certain *type* — a safe rest area, a water-like resource, a post-hazard recovery exit, or a
29 goal-associated region — specified by a semantic pattern rather than a fixed (x, y) waypoint. The
30 place must therefore be *remembered and retrieved by meaning*. This setting opens a methodological
31 question: can we measure *where* memory-based semantic navigation breaks down, not only *whether*
32 it succeeds?

33 We answer this question with a four-stage diagnostic framework that applies to both single agents
34 and multi-agent collectives. The decomposition is Q/R/M/C: **Q**uery formation, **R**etrieval satisfaction,
35 target **M**aterialization, and **C**ompletion after retrieval. The decomposition factorizes per-episode

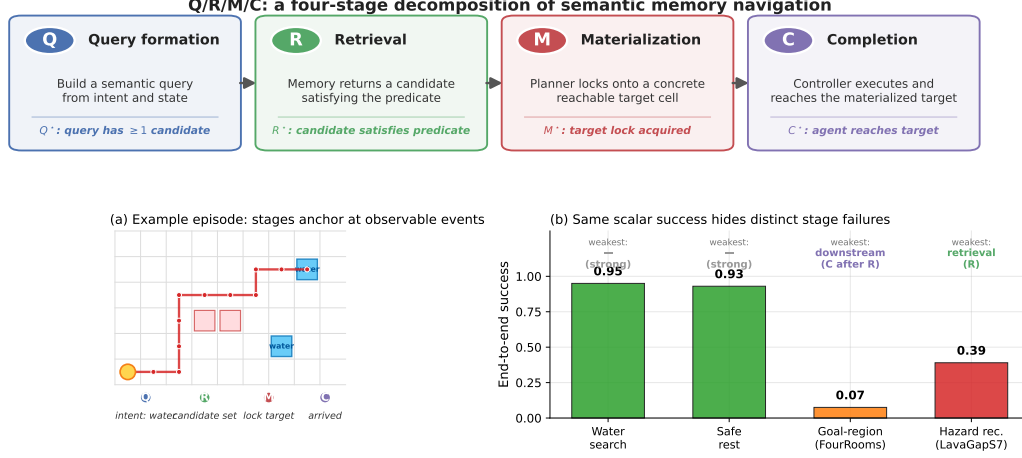


Figure 1: Q/R/M/C decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events Q^* , R^* , M^* , C^* . **(b)** The same scalar “end-to-end success” summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §3.4).

semantic-path completion and, under a conditional stage-Markov property, its factors are estimable from trajectory logs (Proposition 1). The rest of the paper is one continuous argument in three moves.

Move 1 (single agent): stage attribution is non-trivial. Success-only metrics collapse qualitatively different failures into a single scalar. In a 10-seed MiniGrid benchmark, water and rest tasks are strong end-to-end (0.95, 0.93 success), but goal-region rejoin and hazard recovery are substantially harder — and for *different reasons*. Oracle-stage interventions show that hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$), while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without changing end-to-end success). Inspection of the framework’s own logs then sharpens the attribution: the retrieval bottleneck is not candidate *ranking* (precision 96.9% when candidates exist) but candidate *generation* — the memory contains no eligible symbols at 91% of decision points. The single-agent bottleneck is a *memory-accumulation* limit: evidence is observed but never consolidated into queryable structure.

Move 2 (multi-agent): consolidation acquires a rate. When N agents share memory through a broadcast protocol with cadence K , the factorization extends with enlarged conditioning sets (Proposition 2), and K becomes a structural axis: it is, mechanically, a knob on *how fast private memories consolidate into a collective one*. A slope-level claim (Proposition 3) predicts a bottleneck shift: as K decreases, $P(M^*|R^*)$ rises (fresher peer memory enables reliable commitment) and $P(C^*|M^*)$ falls (peers race for the same targets). Both slope signs are confirmed on a 35,640-run sweep ($p < 10^{-4}$), the shift is directly visible as a negative within-cadence correlation between the two links, and the resulting trade-off produces an interior optimum of mean time-to-success at $K = 8$: consolidation has a non-trivial optimal rate.

Move 3 (synthesis): both regimes point at the same object. The single-agent diagnosis (consolidation of evidence into symbols is the limit) and the multi-agent diagnosis (the rate and cost of sharing consolidated memory determine the optimum) converge on one research target: *collective semantic memory* — a shared symbolic structure with explicit merge, trust, and staleness rules, rather than snapshots exchanged on a timer. Section 5 states this program and shows that Q/R/M/C is the measurement instrument it requires: any candidate design is testable by how it moves $P(M^*|R^*)$ and $P(C^*|M^*)$.

What the data do and do not support. A naive reading of the two slopes of Proposition 3 predicts an interior maximum on the conditional product $P(M^*|R^*) \cdot P(C^*|M^*)$. In this environment that prediction fails: the product is monotone in K and approaches the independent-agent boundary. The interior optimum appears instead on mean time-to-success. The gap is reported explicitly — it is the

68 difference between a stage-level operationalisation that can be *wrong* on specific summaries and a
69 stage-agnostic guess that is unfalsifiable.

70 **Why this framing is timely.** Embodied and goal-conditioned agents increasingly retrieve places,
71 objects, or tools by meaning (Andrychowicz et al., 2017; Ghosh et al., 2021; Savva et al., 2019).
72 LLM-agent systems lean on explicit memory layers but evaluate mostly end-to-end (Wang et al., 2023;
73 Shinn et al., 2023; Packer et al., 2023). Multi-agent reinforcement learning likewise increasingly
74 involves communicating populations (Lowe et al., 2017; Foerster et al., 2018), but the question
75 “which agent should know what, and when?” lacks a unified diagnostic vocabulary. Our framework
76 supplies one.

77 Contributions.

- 78 1. **Framework.** A stage-decomposition of memory-based navigation with a probabilistic factoriza-
79 tion whose factors are estimable from standard trajectory logs, for single agents (Proposition 1)
80 and multi-agent collectives (Proposition 2).
- 81 2. **Slope-level claim.** A bottleneck-shift claim (Proposition 3) about how cadence K trades off
82 the M and C stages, verified empirically ($p < 10^{-4}$ on both slopes), with its chronology stated
83 explicitly (§4.3).
- 84 3. **System.** A symbolic-semantic memory stack instantiating the framework, with four communi-
85 cation architectures (independent, shared-bus, central-aggregator, peer-to-peer broadcast).
- 86 4. **Benchmarks.** (a) 10-seed single-agent MiniGrid benchmark with oracle stage interventions; (b)
87 35,640-run multi-agent sweep at $N \in \{3, 5, 8\}$ plus an 8,640-run scale-up at $N = 12$.
- 88 5. **Honest accounting.** Explicit report of what fails: the strict interior-maximum corollary on the
89 conditional product does not hold in this environment; the interior optimum reappears on the
90 efficiency metric.
- 91 6. **Research program.** A data-grounded case that the binding constraint in both regimes is
92 consolidation of evidence into shareable symbolic structure, motivating collective semantic
93 memory as the next target and Q/R/M/C as its evaluation instrument (§5).

94 2 The Q/R/M/C decomposition

95 2.1 Problem formulation and notation

96 The agent operates in a partially observable navigation environment. At each step it receives an
97 observation o_t , maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task
98 intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate
99 is materialized into an actionable target; a local controller executes actions until either the target is
100 reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a
101 function of the chosen candidate’s semantic profile, not of any pre-supplied coordinate.

102 Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva
103 et al., 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class
104 environments to obtain clean and reproducible stage measurements, and use BabyAI only as a
105 portability check. We do not claim to solve LLM-agent memory here; rather, we argue that the
106 same Q/R/M/C decomposition is a useful template for evaluating memory-grounded agent decisions
107 beyond navigation.

108 The central empirical question is therefore not just whether the agent succeeds, but *which stage fails*
109 *first* when it does not. Two related measurement layers are used. **Operational rates** count, per query
110 attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied
111 the semantic predicate, and whether it was materialized into an actionable target; *completion after*
112 *retrieval* is an episode-level indicator. **Nested episode-level events** $Q^* \Rightarrow R^* \Rightarrow M^* \Rightarrow C^*$ are used
113 in the factorization below and audited in §3.2. Full operational definitions are given in Appendix B.

114 **Notation.** The four stages are Q, R, M, C with episode-level starred events $Q^*, R^*, M^*, C^* \in$
115 $\{0, 1\}$, nested by construction. $P(\cdot)$ denotes the population probability; $\hat{P}(\cdot)$ its empirical-frequency
116 estimator. **The letter M always refers to the Materialization stage; the number of targets in**
117 **the multi-agent environment is written T (not M , constraint $T \leq N$, “scarcity” = $N > T$).** Other

118 symbols: $\mathcal{C}_K$ peer-broadcast protocol with cadence K , $\mathcal{M}_{-i}^{(K)}$ peer memory snapshots, $\mathcal{O}_{-i}$ peer
 119 occupancy, $\nu(\cdot)$ freshness functional, $\Phi(K)=P(M^*|R^*;K)\cdot P(C^*|M^*;K)$, t_{succ} mean time-to-
 120 success. Full symbol table in Appendix C.1.

121 2.2 Single-agent factorization

122 Let Y denote overall episode success, and let Q^*, R^*, M^*, C^* denote the nested episode-level
 123 semantic-path events. The framework factorizes *semantic-path completion* rather than raw task
 124 success:

$$Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*.$$

125 For any query q and memory state $\mathcal{M}$,

$$P(C^* = 1 \mid q, \mathcal{M}) = P(Q^* = 1 \mid q, \mathcal{M}) P(R^* = 1 \mid Q^* = 1, q, \mathcal{M}) \\ \cdot P(M^* = 1 \mid R^* = 1, q, \mathcal{M}) P(C^* = 1 \mid M^* = 1, q, \mathcal{M}).$$

126 Overall task success Y can exceed C^* because some episodes succeed without traversing the full
 127 semantic retrieval path.

128 **Assumption 1 (conditional stage-Markov property).** Conditioned on the current stage being
 129 successful, the probability of the next stage depends only on the current successful stage, the current
 130 query, and the current memory state. Formally,

$$P(R^* \mid Q^*, q, \mathcal{M}, \text{history}) = P(R^* \mid Q^*, q, \mathcal{M}),$$

131 and analogously for M^* and C^* .

132 **Assumption 2 (positivity).** Each conditioning event used above has non-zero probability on the
 133 support of the benchmark.

134 **Proposition 1 (single-agent estimability).** Under Assumptions 1–2, the four stage factors are
 135 estimable from trajectory observations and the empirical-frequency estimators are consistent; their
 136 product converges to semantic-path completion. Proof in Appendix A.

137 The mathematical content is elementary — a chain-rule factorization plus consistency of frequency
 138 estimators. The value of the statement is operational: it pins down the exact logging discipline (which
 139 events, conditioned on what) under which the decomposition can be read off standard trajectory logs,
 140 and it is the contract that every experiment in this paper audits against (§3.2).

141 **When Assumption 1 may fail.** Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly
 142 logged at every stage event. Two practically relevant violations remain: adaptive in-episode query
 143 rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the
 144 factorization should be read as a projection onto observable traces rather than as a full structural
 145 identification claim.

146 **Mechanism-level reading.** The same decomposition admits a stage-structured mechanism interpre-
 147 tation. Figure 2 treats stage outcomes as a directed acyclic graph. Under this interpretation, assist
 148 on/off ablations are mechanism-targeted perturbations of materialization and post-retrieval execution
 149 rather than undifferentiated sensitivity analyses; in the multi-agent setting the same graph is replicated
 150 per agent and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the R→M edge
 151 and on $\mathcal{O}_{-i}$ at the M→C edge — exactly the two edges where Proposition 3 will predict the slope
 152 shift.

153 **Relation to prior work.** Closest are (i) cognitive-map and semantic-map frameworks (Tolman,
 154 1948; O’Keefe and Nadel, 1978; Kuipers, 2000; Gupta et al., 2017; Savinov et al., 2018), (ii) goal-
 155 conditioned and successor-representation RL (Andrychowicz et al., 2017; Ghosh et al., 2021; Dayan,
 156 1993; Stachenfeld et al., 2017; Momennejad et al., 2017), (iii) multi-agent communication (Lowe
 157 et al., 2017; Foerster et al., 2018; Sukhbaatar et al., 2016; Foerster et al., 2016) and gossip/consensus
 158 (Boyd et al., 2006; Xiao et al., 2007), and (iv) LLM-agent memory layers evaluated end-to-end (Wang
 159 et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal to all four: it
 160 *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails and lifts the cadence
 161 axis from network protocols to a cognitive-architecture parameter (full discussion in Appendix E.3).

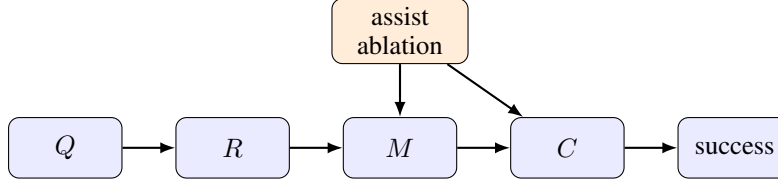


Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the $R \rightarrow M$ and $M \rightarrow C$ edges.

Table 1: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^*)$	$\hat{P}(R^* Q^*)$	$\hat{P}(M^* R^*)$	$\hat{P}(C^* M^*)$	Product
Water	water-cp	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	rest-s2	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	goal-n1	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	hazard-s7	0.40	0.16	1.00	0.56	0.58	0.50	0.16

3 Single agent: stage diagnostics localize failure

3.1 System, tasks, and protocol

Memory stack. The system uses a graph substrate with a semantic retrieval interface on top: observations o_t are tokenized into symbolic scene tokens and semantic tags that update a memory of place records, transitions, episodic traces, and concept/relation records. Planner queries return candidates by a weighted log-score over required, preferred, and penalty tags. The retrieval machinery is reused across all tasks and architectures; full memory schema, update rules, and scoring formula are in Appendix C.

Tasks, baselines, protocol. Four single-agent task families are evaluated (water, safe-rest, goal-region on Empty-6x6 / FourRooms, hazard recovery on LavaGapS7), 10 seeds $\times$ 8 episodes each, $\times$ 2 assist modes. Semantic stack (node, concept, state-conditioned variants) is compared against four baselines: random, graph-only, SPTM-like, learned BC. Reported: end-to-end success, steps, four operational stage rates, assist deltas, weakest stage per task and per method. All means are seed-aggregated; 95% bootstrap CIs use $B=5,000$ resamples; assist comparisons use paired Wilcoxon + Holm-Bonferroni (Appendix F). For the two hardest tasks (FourRooms, LavaGapS7), four oracle regimes (*base*, *oracle R/M/C*) are run — each replaces exactly one stage while leaving the rest of the pipeline unchanged. BabyAI portability check in Appendix D. All single-agent experiments ran on a 10-core ARM CPU, 32 GB RAM (CPU-only); oracle sweep < 1 min wall-clock.

3.2 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 1) — exactly the contract of Proposition 1.

3.3 Success-only summary and what it hides

Water and rest are strong end-to-end tasks (0.95 and 0.93 success; Figure 1b). Goal-region reaches 0.54 success at the task-family level, but the bottleneck sits downstream of candidate selection (§3.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage. Under success-only evaluation the two hard families look like the same kind of

Table 2: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

weakness; the stage view will show they are opposites. Per-task operational rates with 95% CIs are in Appendix C.4.

Baseline comparison. Table 2 compares the semantic stack against random, graph-only, SPTM-like, and learned behavior-cloned baselines. The learned baseline dominates the easiest resource tasks (1.00 on water and rest) and hazard recovery (0.74), while the semantic stack remains strongest on the relation-heavy goal-region family (0.54). The relevant contrast for this paper, however, is not the success column: the semantic stack is the only method family whose decisions expose interpretable Q/R/M/C structure. On the BC baseline the stage events are degenerate for the same structural reason as on the multi-agent learned baseline analysed in §4.5 — a policy without an explicit query/lock interface cannot emit separable stage events, so the diagnostic decomposition has nothing to attach to. End-to-end learning and stage diagnostics answer different questions; this paper is about the second.

3.4 Oracle-stage interventions localize the bottleneck

On goal-region the semantic variants solve Empty-6x6 (1.00 success) but collapse on FourRooms (0.075). For hazard recovery (LavaGapS7), base success 0.39 rises only to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval; goal-region rejoin is downstream-limited (oracle R lifts semantic-path completion to 0.08 without moving end-to-end success). The per-task attribution as a stacked waterfall is in Figure 5 (Appendix C.3). Two tasks identical under success-only evaluation fail at opposite ends of the Q/R/M/C chain.

3.5 The logs refine the diagnosis: a memory-accumulation limit

Inspecting the framework’s own logs sharpens the retrieval attribution. On the 10-seed hazard-recovery run, 349 retrieval calls were logged: only 32 (9.2%) returned at least one candidate, and of these 31 (96.9% precision) selected a semantically satisfied candidate. The bottleneck is therefore not candidate *ranking* but candidate *generation* — the symbolic memory does not contain hazard-recovery-tagged nodes at 91% of decision points. A retrained 2-layer candidate-generation MLP trained on per-state features (test accuracy 0.59 vs majority-class 0.81; Appendix G) confirms that instantaneous state features cannot substitute for the missing structure. The framework’s “retrieval-limited” attribution thus refines to a *memory-accumulation* limit: candidate generation cannot be solved from state features alone, and the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This sets up the multi-agent section: when consolidation becomes a shared process across N agents, it acquires a measurable *rate*.

4 Multi-agent: memory as a shared resource

4.1 Four memory architectures and the cadence axis

Four communication architectures are instantiated; they share the single-agent retrieval substrate but differ in how memory is distributed across agents (Figure 3).

Independent. Each agent maintains a private event store and concept graph. No inter-agent communication mechanism exists. Lower-bound baseline.

Shared bus. All agents publish observations to a single *collective memory* event bus. One shared concept graph is rebuilt from the union of events. Maximally centralized.

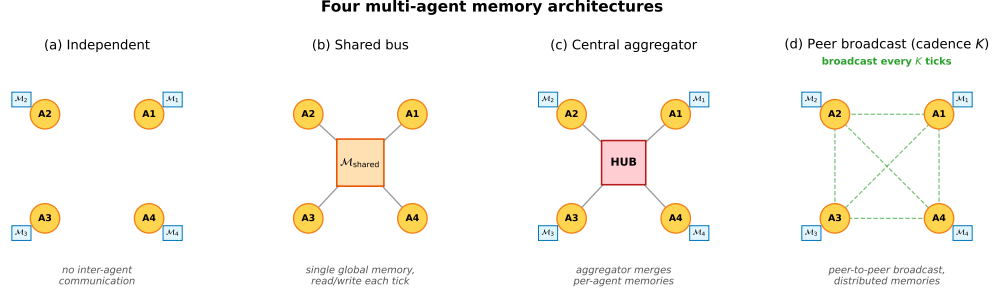


Figure 3: Four multi-agent memory architectures. **(a) Independent:** per-agent private memory, no exchange. **(b) Shared bus:** one global memory, every agent reads/writes each tick. **(c) Central aggregator:** per-agent memories plus a hub that merges them into a consensus report. **(d) Peer-to-peer broadcast:** per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Proposition 3).

Central aggregator (ConsensusLayer). Each agent has its own private concept graph; a central layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report* consumed by all agents.

Peer-to-peer broadcast (cadence K). Each agent has its own concept graph and its own merged view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently merges its own snapshot with the latest snapshot received from each peer. Different agents may hold different merged views; there is no central report.

The first three correspond to the three regimes in the multi-agent communication literature: no exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the architecture for which Propositions 2 and 3 are directly applicable, since the cadence K is a tunable parameter rather than a fixed protocol. Mechanically, K is the rate at which private memories consolidate into a collective view — which is why it is the structural axis of this section.

4.2 Multi-agent factorization

Extend the factorization to a population of N agents that share information through a communication protocol $\mathcal{C}_K$ parameterized by broadcast cadence K . Each agent i has its own memory state $\mathcal{M}_i$ and issues its own query q_i . Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i 's memory at the moment it queries, after absorbing every peer message received under cadence K . Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots accessible to agent i ; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already claimed by the time i acts.

Proposition 2 (multi-agent factorization, estimability). For each agent i ,

$$P(C_i^* | q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^* | q_i, \mathcal{M}_i) P(R_i^* | Q_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)}) \\ \cdot P(M_i^* | R_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)}) P(C_i^* | M_i^*, q_i, \mathcal{O}_{-i}).$$

Under a multi-agent stage-Markov assumption (Appendix B), the four factors are estimable from per-agent trajectory logs and the empirical-frequency estimators are consistent.

When Proposition 2 may fail. The single-agent stage-Markov property absorbs trajectory history into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how* peers arrived there. The factorization is therefore a projection onto observable per-agent traces under that summary. Two specific failure modes arise: (i) inter-agent communication side-channels not logged through $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not summarised by $\mathcal{O}_{-i}$.

4.3 Bottleneck shift

Proposition 3 (bottleneck shift $M \leftrightarrow C$, slope-level claim). Assume scarcity ($N > T$ targets, see §2.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions are posited: (F) memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and the probability that agent i ’s planner locks onto a reachable target is non-decreasing in ν ; (O) the probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases. Write $P(M^*|R^*; K)$ and $P(C^*|M^*; K)$ for the population-level conditional probabilities (averaged over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial P(M^*|R^*; K)}{\partial K^{-1}} > 0, \quad \frac{\partial P(C^*|M^*; K)}{\partial K^{-1}} < 0.$$

Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

Strict interior-maximum corollary (NOT empirically supported). A naive reading of (F) and (O) suggests that the product $\Phi(K) := P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ admits an interior maximum at some $K^* \in \{2, \dots, K_{\max} - 1\}$. In the 35,640-run sweep (§4.5) this corollary does *not* hold: Φ is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and continues to increase toward the independent-agent boundary value $\Phi(K \rightarrow \infty) = 0.605$. The slopes (F) and (O) are real, but the M-side gain in $P(M^*|R^*)$ at small K is not large enough to overcome the C-side cost; the two effects do not produce an interior peak in Φ in this environment. The interior optimum that the framework intuitively predicts does appear, however, on the practically relevant efficiency metric mean time-to-success: this metric has a clear minimum at $K = 8$ (§4.5), with significantly worse performance at both faster and slower cadences. The conditional product Φ captures the chain probability $P(M^* \wedge C^* | R^*)$ but ignores the cost of getting through the chain.

Status and chronology. Assumptions (F) and (O) are intuitive but not derived from a generative model; a formal proof would require, e.g., modelling memory updates as a Poisson-renewal process with rate λ and occupancy as a birth-death process whose rate scales with K^{-1} . Proposition 3 is therefore treated as a slope-level empirical claim verified on data, separating the verified slope conditions from the corollary that does not survive the data. We also state the chronology explicitly: Proposition 3 was formulated *after* an initial 12,960-run sweep conducted under four stage-agnostic hypotheses (Appendix E); its testable slope consequences were then confirmed on the full 35,640-run sweep and, independently, on the $N = 12$ scale-up (Appendix D). It is a post-hoc structural explanation whose predictions were verified on enlarged data, not a pre-registered prediction. A fully formal version is left for future work.

4.4 Sweep protocol

The four architectures are evaluated on a custom multi-agent grid (12×10 cells); each agent must reach a cell tagged `water_source`, choosing actions via a shared memory-driven planner so only the memory layer differs. The sweep crosses $N \in \{3, 5, 8\}$, $T \leq N$, three layouts (symmetric/asymmetric/random), three hazard densities, four architectures, eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds — 35,640 runs in ≈ 22 min on 8 cores. Per-agent per-tick Q/R/M/C indicators are logged (Appendix B); episode-level starred events are the disjunction over ticks. A separate oracle-intervention block on scarcity cases $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$ at five cadences evaluates oracle R (memory returns ground-truth water positions) and oracle M (planner locks onto nearest real water).

4.5 Results: stage decomposition and bottleneck shift

The 35,640-run sweep is reported through four observations, structured to mirror Proposition 3: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$, peaking at $K = 4$ and decaying to noise at $K = 64$; (iii) on mean time-to-success, the resulting trade-off produces a clean interior optimum at $K = 8$; (iv) on the conditional product Φ , the same trade-off produces a monotone trend rather than an interior peak — the corollary that does not

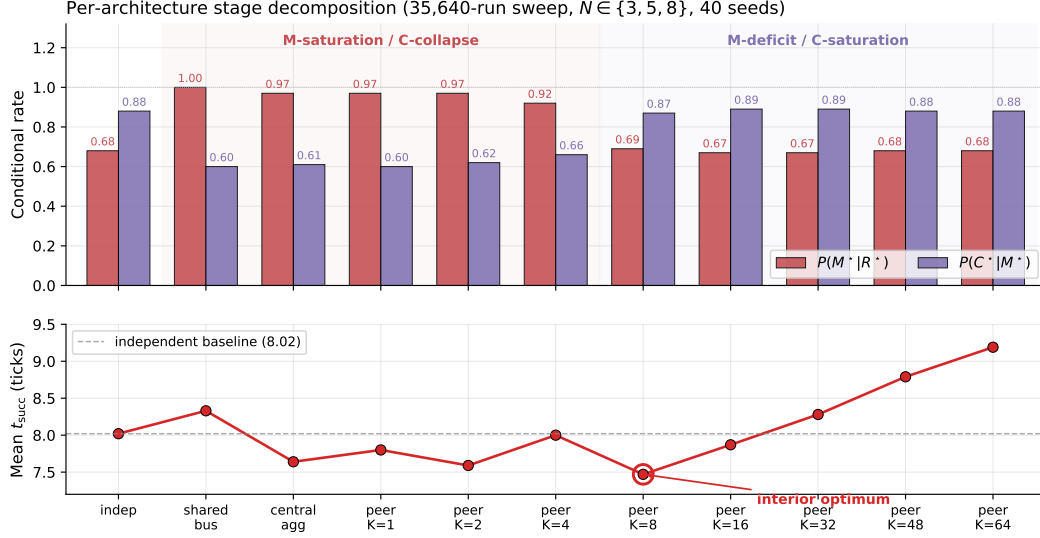


Figure 4: Per-architecture conditional rates (top) and mean time-to-success (bottom) from the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. Mean time-to-success has a clean interior minimum at $K=8$ (7.47 ticks), below the independent-agent baseline (8.02).

307 survive the data. A learned-communication baseline and a multi-agent oracle-intervention loop close
 308 the subsection.

309 4.5.1 Conditional decomposition reveals where each architecture fails.

310 Figure 4 shows per-architecture conditional $Q/R/M/C$ rates; full numerics with bootstrap CIs are in
 311 Appendix E.1, Table 5. Retrieval conditional $P(R^*|Q^*)$ is saturated; the differentiating signal lives
 312 at M and C. Shared-bus, central-aggregator, and fast-broadcast peer ($K \leq 4$) have high $P(M^*|R^*)$
 313 but low $P(C^*|M^*)$; independent and slow-broadcast peer ($K \geq 16$) show the reverse — the $M \leftrightarrow C$
 314 trade-off predicted by Proposition 3.

315 4.5.2 Bottleneck shift and interior efficiency optimum.

316 Across all peer runs, Spearman of $P(M^*|R^*)$ vs K is -0.608 and of $P(C^*|M^*)$ vs K is $+0.420$
 317 (both $p < 10^{-4}$), matching the slope conditions of Proposition 3. The stronger empirical signa-
 318 ture is at the per-configuration level: Pearson of $P(M^*|R^*)$ vs $P(C^*|M^*)$ within a fixed K is
 319 $-0.43, -0.54, -0.61, -0.19, -0.05, -0.04, -0.02, -0.02$ at $K=1, 2, 4, 8, 16, 32, 48, 64$, peaking
 320 at $K=4$ and decaying to noise by $K=64$ (Figure 8, Appendix E.1). Configurations where M is
 321 high systematically have low C: the bottleneck migrates between the two stages within the same
 322 ($N, T, \text{layout, hazard, seed}$) cell as K varies. On the efficiency metric, mean t_{succ} has a U-shape with
 323 interior minimum at $K=8$ (7.47 ticks vs 8.02 for the independent baseline; paired-bootstrap CIs
 324 in Appendix E.1). Scaling to $N=12$ sharpens the signature: the peak Pearson rises to -0.70 and
 325 migrates outward to $K=8$, consistent with stronger occupancy coupling at higher N delaying the
 326 cost-regime transition (Appendix D, Figure 6).

327 **Honest limit on the conditional product.** The conditional product $\Phi = P(M^*|R^*) \cdot P(C^*|M^*)$
 328 does *not* have an interior peak in the tested range: the mean-of-products estimator increases mono-
 329 tonically from 0.569 at $K=4$ to 0.603 at $K=64$ and converges to the independent-agent boundary
 330 0.605. The slopes of Proposition 3 are supported; the additional interior-maximum corollary is not,
 331 in this environment. Jensen-gap detail and POM-MOP comparison in Appendix E.1.

332 **Learned-communication baseline: stage events are structurally degenerate even at competitive**
 333 **success.** A CommNet-style policy (Sukhbaatar et al., 2016) trained with PPO on the same scenario

($N=3$, $T=2$) reaches 0.667 held-out success ($n=100$ seeds) — matching the symbolic peer architecture at $K=8$ (0.65). Despite this, the Q/R/M/C profile remains structurally degenerate: $Q^*=1.00$ saturated by construction; $R^*=0.997$, $M^*=0.997$ collapsed together ($P(M^*|R^*)=1.00$). This is the verified structural claim: a real-valued comm vector is never informatively empty (Q saturates) and lacks an explicit target-lock interface (R/M merge), regardless of training budget. The multi-agent counterpart of the single-agent learned-BC analysis (§3.3). Full architecture, PPO hyperparameters, and side-by-side REINFORCE vs PPO comparison in Appendix G.

Multi-agent oracle interventions close the diagnose→intervene→verify loop. On the scarcity cases, mean t_{succ} at small/intermediate cadence ($K \in \{1, 2, 4\}$) drops from ~ 10 – 11 in baseline to ~ 7.1 under either oracle R or oracle M; at slow cadence ($K=16$) the baseline is already within ~ 0.8 of the oracle floor (Table 6 in Appendix E.4). The cadence-induced excess time at small/intermediate K is attributable to the $M \leftrightarrow C$ transition; at large K the bottleneck has moved into solo exploration.

Bridge. The multi-agent block thus delivers its half of the argument. The cadence K is, mechanically, the rate at which private memories consolidate into a collective view; the data show that this rate has a non-trivial interior optimum ($K=8$ on t_{succ}), that its cost channel is occupancy coupling, and that both effects are visible stage-by-stage on the same Q/R/M/C chain used for single agents. Consolidation is not only the single-agent limit — in a collective it is a tunable, measurable design axis.

5 Synthesis: toward collective semantic memory

The single-agent and multi-agent halves of this paper run on different substrates, with different agents and different questions, yet they return the same answer. The single-agent diagnosis localized the hardest failure not to retrieval mechanics but to *consolidation*: candidate ranking is near-optimal (96.9% precision when candidates exist), but candidate generation is starved (91% empty candidate sets), and instantaneous state features cannot substitute for the missing structure (§3.5). The multi-agent diagnosis showed that when consolidation becomes a shared process it acquires a *rate* (the cadence K) with a benefit channel at the $R \rightarrow M$ edge and a cost channel at the $M \rightarrow C$ edge, whose trade-off produces an interior optimum neither fully centralized nor fully independent architectures can reach. The binding constraint in both regimes is the consolidation of evidence into shared symbolic structure.

We therefore propose *collective semantic memory* — a shared symbolic structure with explicit **merge**, **trust**, and **staleness** rules — as the research target the data select. The architectures of §4.1 are degenerate corners of this design space (shared bus merges everything instantly and pays full occupancy cost; independent never merges and pays full staleness; peer-broadcast exposes exactly one design parameter, the rate K). The framework built in this paper is the measurement instrument any candidate design requires: a good consolidation rule must raise $P(M^*|R^*)$ without collapsing $P(C^*|M^*)$ — that is, reach the upper-right region of Figure 8 that no fixed cadence reaches — and reduce the empty-candidate-set fraction of §3.5. Concrete open problems (adaptive cadence, selective consolidation, intent-weighted trust, substrate generality) are listed in Appendix E.2.

6 Limitations and future work

Two environments. Single-agent runs on MiniGrid/BabyAI; multi-agent on a custom 12×10 grid. Identical operational Q^* , R^* , M^* , C^* definitions recover interpretable stage profiles on both substrates, with the same chain ordering and weakest-stage attribution. A multi-agent MiniGrid wrapper unifying the two is the next portability step (Appendix E.2, item 4).

Proposition 3 at slope level only. The two slope conditions hold ($p < 10^{-4}$); the strict interior-maximum corollary on Φ does not (§4.3). A version of Proposition 3 stated directly in time-cost would close this gap. Proof under a generative model (Poisson-renewal memory, birth-death occupancy) is future work.

Estimability under hidden state. With unlogged in-episode query rewrites or peer side-channels, Propositions 1 and 2 are projections onto logged traces, not full identification.

Learned baselines. The CommNet PPO baseline reaches 0.667 success on the scarcity scenario — competitive with symbolic peer at $K=8$ (0.65) — and confirms Q-saturation ($Q^*=1.00$) and R/M collapse ($P(M^*|R^*)=1.00$) at competitive success. The structural prediction now holds verified, not asserted; ablating on architectures with explicit symbolic target-lock channels is left for future work.

Scale. Multi-agent sweep bounded at $N=12$ (Appendix D; signature sharpens, peak Pearson -0.70 at $K=8$, interior t_{succ} minimum 4.77). Denser scaling at $N \geq 16$ and continuous-state environments are future work.

7 Broader impacts

The paper proposes diagnostic evaluation for memory-based agents in simulated navigation. Its positive impact is methodological: making memory-based failures more measurable across architectures, and making the cadence axis of distributed multi-agent memory tunable on an empirically grounded criterion. Potential negative impacts are indirect and would arise if similar diagnostics were used to improve agents in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks) without adequate safeguards. The current work is benchmark-level infrastructure, not a deployment-ready system.

8 Conclusion

We proposed Q/R/M/C, a four-stage decomposition of memory-based navigation whose factors are estimable from standard trajectory logs, and used it to run a single continuous argument: stage diagnostics localize failures that success-only evaluation cannot (single agent); when memory is shared, the exchange cadence becomes a measurable structural axis with an interior optimum (multi-agent); and both measurements localize the binding constraint to the same mechanism — the consolidation of trajectory evidence into persistent, shareable symbolic structure. Collective semantic memory — a shared symbolic structure with explicit merge, trust, and staleness rules — is therefore not a speculative extension but the research target that the data select, and the framework presented here is the instrument by which candidate designs for it can be evaluated.

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453 A Formal proofs

454 A.1 Proof of Proposition 1

455 For each episode $i \in \{1, \dots, n\}$ let $Q_i^*, R_i^*, M_i^*, C_i^*$ denote the nested episode-level indicators. By
456 construction these are measurable functions of the runtime trajectory log and satisfy $C_i^* \Rightarrow M_i^* \Rightarrow$
457 $R_i^* \Rightarrow Q_i^*$. The empirical estimators

$$\hat{p}_Q = \frac{1}{n} \sum_i Q_i^*, \quad \hat{p}_{R|Q} = \frac{\sum_i R_i^*}{\sum_i Q_i^*}, \quad \hat{p}_{M|R} = \frac{\sum_i M_i^*}{\sum_i R_i^*}, \quad \hat{p}_{C|M} = \frac{\sum_i C_i^*}{\sum_i M_i^*}$$

458 are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product
459 converges to semantic-path completion. $\square$

460 A.2 Proof of Proposition 2

461 Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the
462 proof of Proposition 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on
463 a measurable event. The qualifications stated in §4.2 (“When Proposition 2 may fail”) describe two
464 specific situations in which the enlarged stage-Markov property is violated. $\square$

465 A.3 Justification of Proposition 3

466 No full proof of Proposition 3 is given here; the slope-level argument under (F) and (O) is recorded
467 below.

468 *Slope (a), under (F).* As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on
469 a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν .
470 Therefore $\partial P(M^*|R^*; K)/\partial K^{-1} > 0$.

471 *Slope (b), under (O).* As K decreases, peers receive updated memory simultaneously and con-
 472 verge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore
 473 $\partial P(C^*|M^*; K)/\partial K^{-1} < 0$.

474 *Interior maximum corollary fails in our data.* A naive reading would predict that $\Phi(K) =$
 475 $P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ admits an interior maximum. In the 35,640-run sweep Φ is monotoni-
 476 cally increasing over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary
 477 value $\Phi_\infty = 0.605$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two opposing terms;
 478 in our environment the M-side gain at small K is not large enough to flip the sign of the sum. A
 479 formal version of the corollary would require quantitative bounds on the relative magnitudes, e.g. via
 480 a generative model in which freshness $\nu(K)$ saturates at a known rate and occupancy scales with
 481 K^{-1} . That is left to future work; the empirical truth is reported as found: slope conditions verified;
 482 interior peak on Φ not found. $\square$

483 B Operational vs. nested stage estimators

484 **Operational diagnostic rates.** The main benchmark logs per-query and per-episode diagnos-
 485 tic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized,
 486 completion after retrieval). These are operational rates, not direct multiplicative factors.

487 **Nested stage estimators for factorization.** For the factorization, episode-level nested events
 488 from the first failure stage recorded in the runtime taxonomy are used: Q^*, R^*, M^*, C^* , nested by
 489 construction. The consistency audit in §3.2 reports $\hat{P}(Q^*), \hat{P}(R^*|Q^*), \hat{P}(M^*|R^*), \hat{P}(C^*|M^*)$,
 490 their product, and its gap to semantic-path completion.

491 B.1 Multi-agent operational definitions

492 For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let
 493 $\text{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t , $\text{Locked}_i(t)$ be the
 494 agent’s planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

495 **Per-tick events.** $Q_i(t) = \mathbf{1}[\text{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$);
 496 $M_i(t) = \mathbf{1}[\text{Locked}_i(t) \neq \emptyset \wedge \min_w \|\text{Locked}_i(t) - w\| \leq \epsilon]$.

497 **Episode-level starred events.** $Q_i^* = \bigvee_t Q_i(t)$; analogously for R^*, M^* ; $C_i^* =$
 498 $\mathbf{1}[\text{agent } i \text{ reached a target}]$.

499 **Aggregation.** Per-run rates are averaged over the N agents; per-configuration rates are averaged
 500 over the 40 seeds. Conditional rates $P(R^*|Q^*), P(M^*|R^*), P(C^*|M^*)$ are computed as ratios of
 501 episode-level counts within each configuration before averaging across configurations.

502 C Memory and task schema

503 C.1 Full notation table

504 Symbols are fixed throughout the paper.

505 **Stages.** R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion after
 506 retrieval. They name both the stage and (in expressions like “the M link”) the corresponding edge
 507 of the chain.

508 **Events.** $R^*, M^*, C^* \in \{0, 1\}$ — episode-level starred events recording whether each stage succeeded
 509 (Appendix B). Nested by construction: $Q^*=1 \Leftarrow R^*=1 \Leftarrow M^*=1 \Leftarrow C^*=1$.

510 **Prob.** $\mathbb{P}(\cdot)$ — population-level probability (averaged over the seed distribution and, in the multi-agent
 511 **a-** case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

512 **bil-** q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed C^*).

513 **stake-** N — number of agents in the population. T — number of targets in the environment (constraint
 514 **gent** $T \leq N$; “scarcity” means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i ’s private memory. q_i —

515 **a-** agent i ’s query.
gent.

516 $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K ticks each agent
517 *planning* broadcasts a memory snapshot, §4.1). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots accessible to
518 agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already claimed by
519 other agents by the time i acts (the *occupancy coupling* channel).

520 *Slope*($\mathcal{M}_{-i}^{(K)}$) — a freshness functional on peer memory; non-decreasing pointwise as K decreases
521 *vari*-assumption (F) of Proposition 3). The occupancy slope is assumption (O).

522 *chain*(K) := $P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ — the conditional product, the chain probability
523 *margin*($M^* \wedge C^* | R^*$). POM (product-of-means) and MOP (mean-of-products) are two estimators of
524 *status*; they differ by $\text{Cov}(P(M^*|R^*), P(C^*|M^*))$, which is the Jensen gap (§4.5). t_{succ} — mean
525 *time*-to-success in environment ticks.

526 C.2 Memory stack and task patterns

527 The memory stack stores symbolic place records (per-place tag confidences, counts, visitation,
528 freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates),
529 episodic traces (intent + state at each visit), and concept/relation records. At each step, the tokenizer
530 emits (z_t, τ_t, e_t) from o_t . Per-place tag tables use a Dirichlet-multinomial update; transitions use
531 exponentially smoothed statistics; episodic records attach the active intent and agent state. A planner
532 query is answered by retrieving candidate places whose semantic profiles match the current task intent
533 and state, via a weighted log-score $s(p | q_t, s_t) = \sum_{\tau \in q^{\text{req}}} \log P(\tau|p) + \alpha \sum_{\tau \in q^{\text{pref}}} \log P(\tau|p) -$
534 $\beta \sum_{\tau \in q^{\text{pen}}} \log P(\tau|p) + \gamma u_{\text{epi}}(p, s_t)$ over required, preferred, and penalty tags. The four task
535 families are defined by semantic place patterns: water search (target {water}), safe rest ({rest} with
536 state-conditioning), goal-region rejoin ({goal-region}), hazard recovery ({post-hazard-exit}).

537 C.3 Oracle-stage intervention waterfall

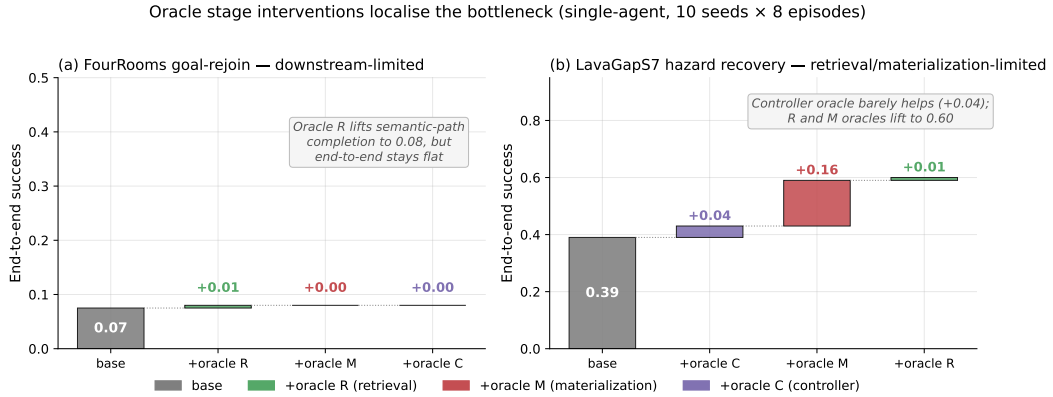


Figure 5: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). (a) FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. (b) LavaGapS7 hazard recovery: oracle controller adds only +0.04, but oracle materialization adds +0.16 and oracle retrieval adds a further +0.01 to reach 0.60. The framework localises the bottleneck stage-by-stage.

538 C.4 Per-task operational rates with 95% CIs

539 D Portability and scale-up

540 D.1 BabyAI portability transfer

541 A 5-seed portability check on BabyAI-GoTo0bjMaze-v0 (8 episodes/seed) confirms that the same
542 Q/R/M/C events are recoverable from BabyAI traces and that the stage-rate ordering matches the
543 MiniGrid benchmark within bootstrap CIs.

Table 3: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	water-cp	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	rest-s2	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	goal-n1	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	hazard-s7	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

544 D.2 Multi-agent scale-up at $N = 12$

545 To check that the multi-agent results of §4.5 are not an artefact of the $N \leq 8$ range, an additional
546 8,640-run sweep was run at $N = 12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities,
547 eight peer cadences, and 40 seeds. Wall-clock ≈ 12 minutes on 8 cores. The conditional-rate
548 signature of §4.5 persists and intensifies (Table 4).

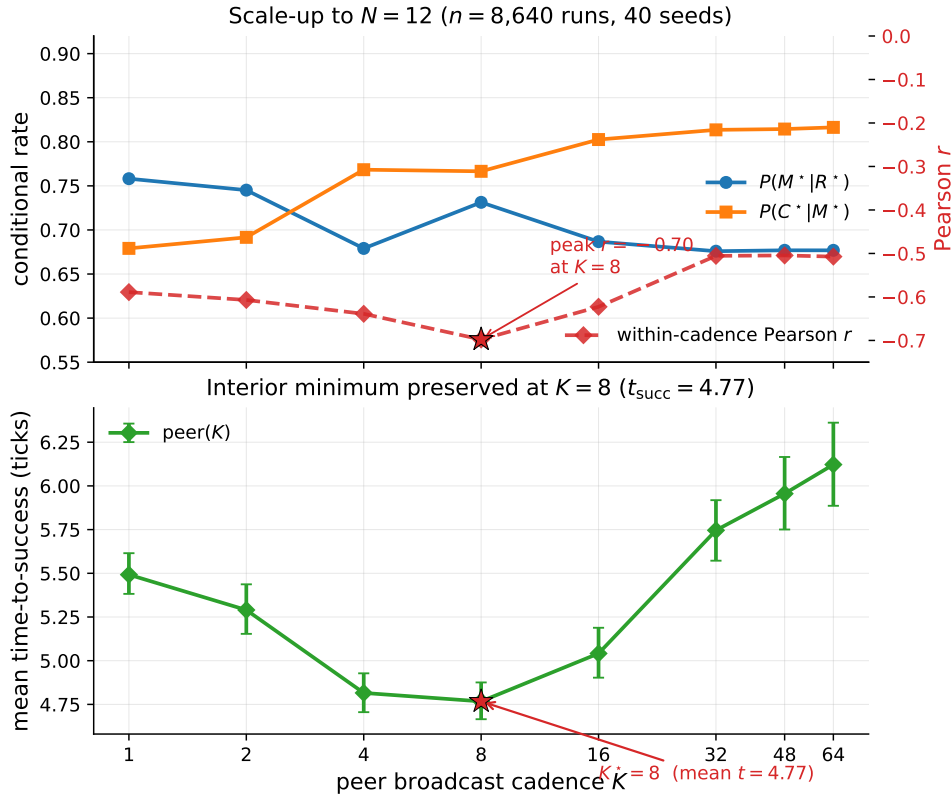


Figure 6: $N = 12$ scale-up bottleneck signature ($n = 8,640$ runs, 40 seeds). Top: both conditional rates change monotonically in K as predicted; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K = 8$. Bottom: mean time-to-success retains an interior minimum at $K = 8$ (4.77 ticks). The bottleneck-shift peak migrates outward from $K = 4$ at $N \leq 8$ to $K = 8$ at $N = 12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

549 E Extended diagnostics

550 E.1 Conditional-product detail and the Jensen gap

551 The conditional product $\Phi = P(M^*|R^*) \cdot P(C^*|M^*)$ in the 35,640-run sweep is monotonically
552 increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary

Table 4: $N = 12$ scale-up sweep numerical detail (peer architecture). Slope conditions hold: Spearman of $P(M^*|R^*)$ with K is -0.141 ($p < 10^{-39}$), and of $P(C^*|M^*)$ with K is $+0.304$ ($p < 10^{-184}$).

K	$P(M^* R^*)$	$P(C^* M^*)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	4.77	-0.70
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

Table 5: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^*|Q^*)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold:** MOP-best of the peer cadences. Mean t_{succ} (last column) is the practically relevant efficiency measure; its interior minimum at $K = 8$ is the predicted-shape result that survives the data.

Architecture	$P(R^* Q^*)$	$P(M^* R^*)$	$P(C^* M^*)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	+0.00	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K = 1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K = 2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K = 4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K = 8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	7.47
peer ($K = 16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K = 32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K = 48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K = 64$)	0.99	0.68	0.88	0.599	0.603 [0.594, 0.611]	-0.02	9.19

553 $\Phi(K \rightarrow \infty) = 0.605$. Paired bootstrap: $\Delta(K=32-K=16) = +0.008$ (CI [+0.006, +0.010]),
554 $\Delta(K=48-K=16) = +0.011$, $\Delta(K=64-K=16) = +0.012$; independent is significantly above
555 every peer K except $K=64$ (borderline).

556 The negative within-cadence covariance documented in §4.5 creates the Jensen-inequality gap
557 between Φ and the marginal product-of-means: $E[X] \cdot E[Y] > E[X \cdot Y]$ when $\text{Cov}(X, Y) < 0$. The
558 mean-of-products is reported as the primary summary because it absorbs the covariance correctly.
559 Figure 7 shows the K-sweep view of both conditional rates and mean t_{succ} ; Figure 8 shows the
560 per-configuration scatter that the negative covariance summarises.

561 E.2 Open problems for the collective-semantic-memory program

562 Four concrete questions follow directly from the measurements of §4.5 and the synthesis of §5. **(1)**
563 **Adaptive cadence.** Proposition 3 assumes a fixed K ; the bottleneck-shift signature suggests an
564 online controller K_t that broadcasts when the (estimated) M-side gain exceeds the C-side occupancy
565 cost. **(2) Selective consolidation.** Snapshot exchange merges everything; merge rules that prioritize
566 evidence about under-represented semantic tags attack the candidate-generation starvation of §3.5
567 directly. **(3) Trust-weighted merging under occupancy.** The central aggregator already performs
568 a trust-weighted merge but pays the C-collapse cost; decentralized trust rules that account for
569 peer *intent* (not only peer evidence) could decouple freshness from target racing. **(4) Substrate**
570 **generality.** The decomposition transfers across MiniGrid, BabyAI, and the custom grid with identical
571 operational definitions; a multi-agent MiniGrid wrapper and a continuous-state environment are the
572 next portability steps.

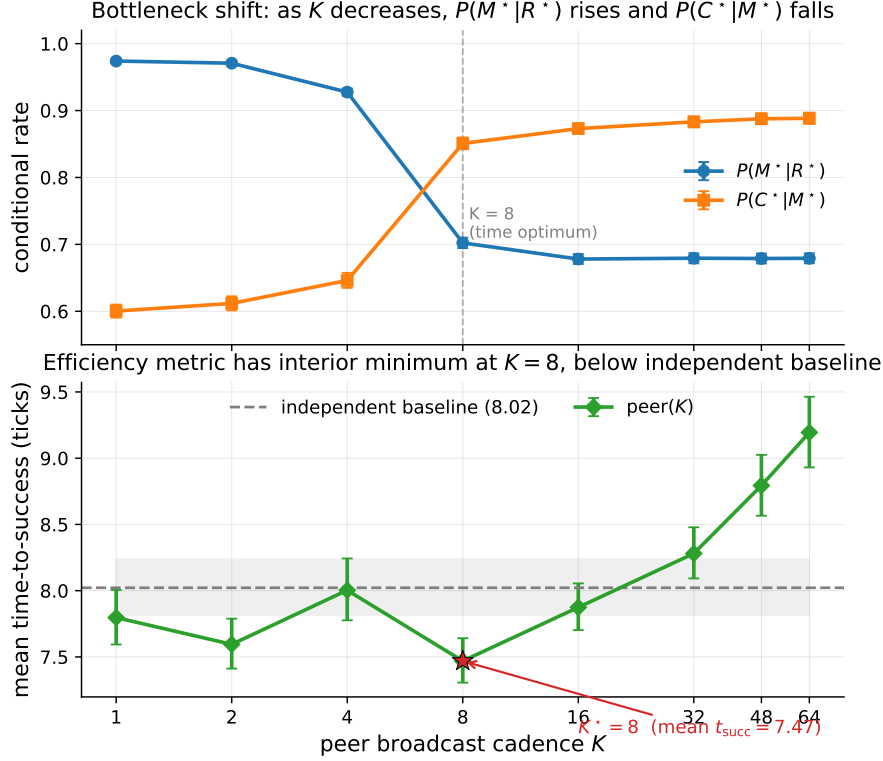


Figure 7: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^*|R^*)$ falls and $P(C^*|M^*)$ rises monotonically in K ($p < 10^{-4}$, both Spearman). Bottom: mean time-to-success has an interior minimum at $K = 8$ (7.47 ticks), below the independent-agent baseline (8.02, dashed). Error bars and shaded band are 95% bootstrap CIs.

573 E.3 Extended related work

574 Classical cognitive-map perspectives motivate explicit place abstractions (Tolman, 1948; O’Keefe
575 and Nadel, 1978; Kuipers, 2000); modern embodied AI combines mapping and planning (Gupta
576 et al., 2017) or exposes topological memory (SPTM, Savinov et al., 2018); goal-conditioned RL
577 conditions on explicit goals or embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021), with
578 successor-representation work providing predictive structure (Dayan, 1993; Stachenfeld et al., 2017;
579 Momennejad et al., 2017). Our setting differs in that the target is a semantic pattern retrieved from
580 memory rather than a coordinate. In multi-agent RL, CTDE dominates (Lowe et al., 2017; Foerster
581 et al., 2018) and learned communication adds a per-step exchange variable (Sukhbaatar et al., 2016;
582 Foerster et al., 2016); gossip and consensus protocols (Boyd et al., 2006; Xiao et al., 2007) are the
583 closest network-level analogue of our cadence axis, lifted here to the cognitive-architecture level.
584 LLM-agent systems lean on explicit memory but evaluate end-to-end (Wang et al., 2023; Shinn et al.,
585 2023; Packer et al., 2023). Our framework is orthogonal: rather than learning a communication
586 policy end-to-end, it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails,
587 and yields a slope-level prediction about cadence that is testable on logs.

588 E.4 Multi-agent oracle-intervention numerics

589 E.5 Stage-agnostic cadence hypotheses and what the framework changes

590 A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run
591 sweep in this paper) was originally planned with four operational hypotheses formulated *before*
592 Proposition 3. They are retained here because they motivated the eventual framework. The framework
593 was formulated after the sweep was complete; it is not claimed to have predicted the sweep numbers
594 a priori (see “Status and chronology” in §4.3).

Bottleneck-shift signature: negative within-cadence correlation peaks at $K = 4$ and decays at $K = 64$

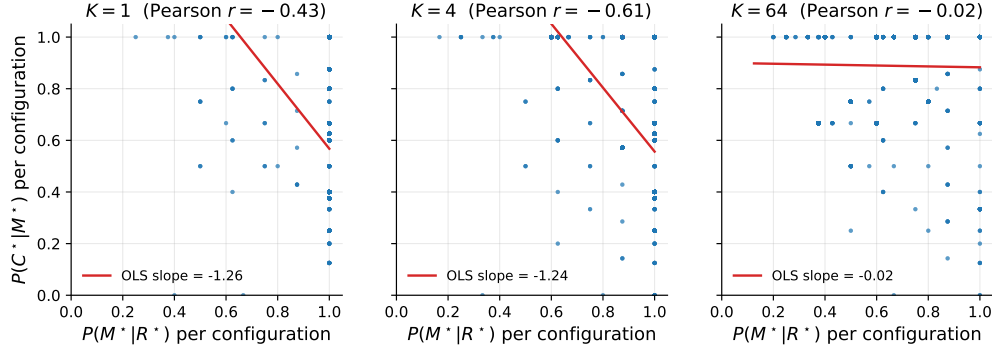


Figure 8: Per-configuration scatter of $P(M^*|R^*)$ vs. $P(C^*|M^*)$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

Table 6: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success averaged across $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, two layouts (asymmetric, random), 15 seeds.

K	Baseline	Oracle R	Oracle M	Gap to oracle
1	10.33	7.22	7.08	-3.11
4	10.97	7.22	7.08	-3.75
16	7.98	7.22	7.08	-0.76

595 Original hypotheses (pre-framework).

596 **H1.** An interior $K^* \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, T, layout)
597 cells.

598 **H2.** K^* grows monotonically with scarcity $\rho = N/T$.

599 **H3.** When $\rho \leq 1$, $K^* = 1$.

600 **H4.** Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success,
601 success rate).

602 **Stand-alone result on the original 20-seed sweep.** The practical claim was supported: peer at the
603 empirically best K was faster than centralized in the scarcity regime ($p = 0.044$, Mann-Whitney).
604 Among the four structural hypotheses, only H4 was clearly supported in the random-layout subset.
605 H1 held in 7 of 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9
606 admissible cells. The 35,640-run extended sweep used elsewhere in the paper produces matching
607 stand-alone numbers within bootstrap uncertainty.

608 **What the framework changes.** The hypotheses H1–H4 are stage-agnostic: each ties K^* to scarcity
609 ρ without specifying mechanism. Proposition 3, formulated after the sweep, ties the slopes of
610 $P(M^*|R^*)$ and $P(C^*|M^*)$ to cadence; the slope conditions are testable. The data verify both signs
611 ($p < 10^{-4}$), but the corollary about an interior maximum on the conditional product fails. The interior
612 optimum holds on the efficiency metric mean time-to-success.

613 F Assist on/off analysis

614 Across all four single-agent task families, top-line success is essentially unchanged under assist on/off
615 ($\Delta \leq 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion
616 by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and
617 post-retrieval execution.

G Retrained retriever and CommNet baseline

G.1 Retrained candidate-generation MLP (§3.5)

Dataset extraction. From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0, method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113 positives (13.9%).

Split. By seed: train {2, 3, 4, 5, 6, 7} (505 examples, 9.7% positive), val {8, 9} (170, 22.4% positive), test {10, 11} (139, 18.7% positive).

Architecture. $18 \rightarrow 32 \rightarrow 32 \rightarrow 1$, ReLU activations, BCE with `pos_weight` ≈ 9.3 for class imbalance, Adam lr = $2 \cdot 10^{-3}$ with `weight_decay` = 10^{-4} , batch 64, 200 epochs, best by val loss.

Results.

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall = 0.70	0.171	0.278	0.275

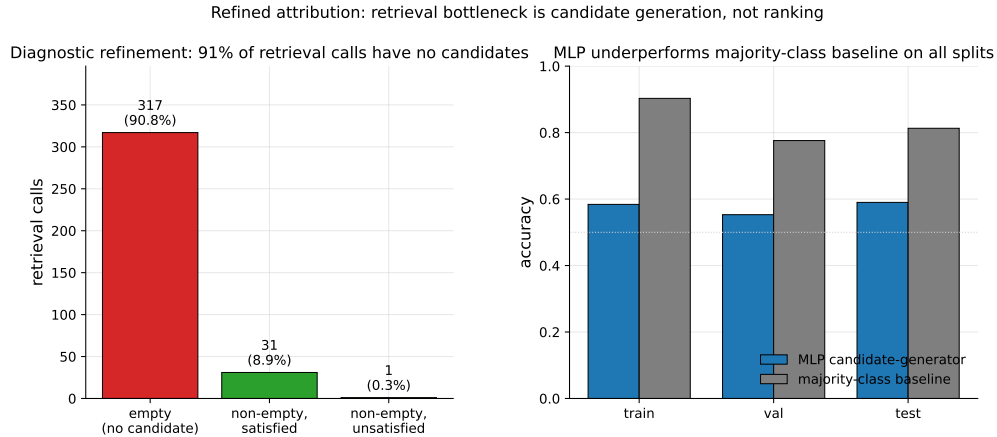


Figure 9: Refined attribution diagnostic. *Left*: 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right*: the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

The MLP underperforms majority-class on accuracy across all splits and achieves only moderate precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility downstream. This is the negative-but-informative finding referenced in §3.5: the framework’s “retrieval-limited” attribution refines to “memory-accumulation-limited” rather than “ranking-limited”.

G.2 CommNet-style learned communication baseline (§4.5)

Policy architecture. Per-agent observation is 4-dim direction one-hot + $5 \times 5 \times 5$ local cell-type window (125) + 2-dim normalised position = 131. Shared-weights encoder $131 \rightarrow 64 \rightarrow 64$ (ReLU), 16-dim comm projection, peer mean-pool (excluding self), combine $80 \rightarrow 64$ (ReLU), actor head $64 \rightarrow 4$ (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \rightarrow 1$.

640 **Training.** REINFORCE with value-baseline; loss = $-\text{mean}(\log \pi(a) \cdot (R-V)) + 0.5 \cdot \text{MSE}(V, R) -$
641 $0.01 \cdot H[\pi]$; Adam lr = $3 \cdot 10^{-4}$, gradient clipping 1.0. Reward shaping: +1.0 on first success per
642 agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200 environment seeds,
643 wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

644 **Training curve.** Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training;
645 no monotonic improvement after ~ 200 episodes. The policy converges to a local optimum where
646 one of three agents finds water by chance.

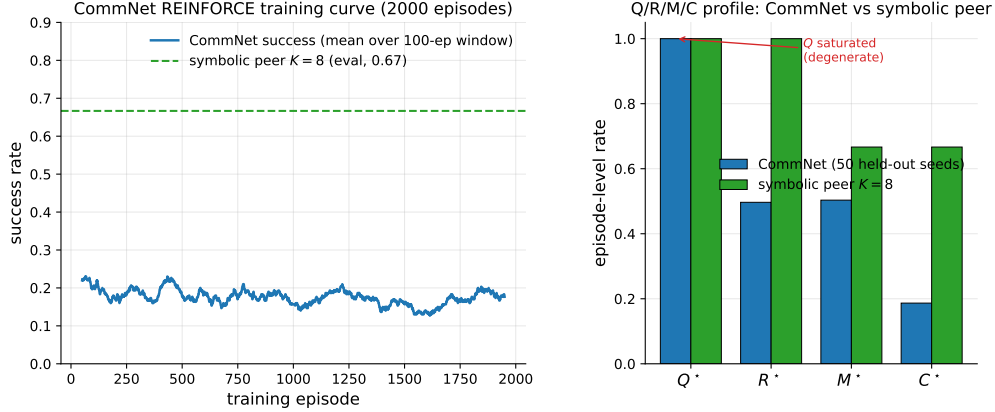


Figure 10: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K=8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K=8$. Q^* is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

647 **Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).**

Quantity	Value
Q^* rate	1.00 (saturated)
R^* rate	0.50
M^* rate	0.50 (collapses onto R)
C^* rate	0.19
$P(R Q)$	0.50
$P(M R)$	1.00
$P(C M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

648 Compare to the symbolic peer architecture at $K=8$: $P(M^*|R^*)=0.69$, $P(C^*|M^*)=0.87$, success
649 ~ 0.65 .

650 **PPO variant: structural degeneracy verified at competitive success.** A PPO-trained variant
651 of the same CommNet architecture (clip $\epsilon=0.2$, GAE $\lambda=0.95$, $\gamma=0.99$, 4 PPO epochs per roll-
652 out, 64 rollouts per update, 200 updates, ≈ 96 minutes on a single CPU core; full setup in
653 `experiments/commnet_ppo_baseline/train_ppo_commnet.py`) reaches 0.667 success on 100
654 held-out seeds — competitive with the symbolic peer architecture at $K=8$ (0.65).

655 The PPO result converts the structural argument from an assertion (“training budget would not change
656 this”) to a measurement: Q^* remains saturated at 1.00, $P(M^*|R^*)$ remains 1.00 (R/M collapse), even
657 at competitive success. The Q stage is degenerate by construction in any policy whose communication
658 channel is a real-valued vector with a learned linear projection — the channel is never informatively
659 empty, regardless of training budget — and without an explicit symbolic target-lock interface the M
660 event cannot separate from R. Only architectures with an explicit *query/lock* interface (the symbolic
661 stack of §4.1) expose stage-separable Q/R/M/C events.

Table 7: REINFORCE vs PPO CommNet vs symbolic peer ($K=8$) on the scarcity scenario ($N=3$, $T=2$, asymmetric, hazard 0.05). PPO is competitive on success rate, yet still exhibits Q-saturation and R/M collapse — the diagnostic claim verified, not asserted.

Metric	CommNet REINFORCE	CommNet PPO	Symbolic peer ($K=8$)
Success rate	0.19	0.667	0.65
Q^* rate	1.00	1.00	0.99
R^* rate	0.50	0.997	0.69
M^* rate	0.50	0.997	0.69
$P(M^* R^*)$	1.00	1.00	0.69
$P(C^* M^*)$	0.42	0.67	0.87
R/M separated?	no	no	yes

662 H Reproducibility commands

663 An anonymized public repository accompanies this submission. The main paper artifacts were
664 generated from the following scripts; PYTHONPATH=. and a Python 3 virtual environment with numpy,
665 scipy, pandas, matplotlib, imageio, gymnasium, and minigrid are assumed throughout.

666 One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)

667 bashscripts/reproduce_paper.sh

668 Single-agent (§3.2–§3.4)

- 669 • article benchmark and learned-baseline suite: scripts/benchmark_symbolic_memory_
- 670 article.py
- 671 • article analysis and figure generation: scripts/analyze_symbolic_memory_article.py
- 672 • synthetic factorization sanity check: scripts/validate_qrmc_factorization.py
- 673 • oracle-stage intervention benchmark: scripts/benchmark_oracle_stage_interventions.
- 674 py
- 675 • BabyAI transfer benchmark: scripts/compare_sonline_babyai.py

676 Multi-agent (§4.5)

- 677 • Main 25,920-run cadence sweep (40 seeds, $K \leq 16$, ≈ 18 min): pythonexperiments/
- 678 big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_
- 679 experiment_qrmc_40
- 680 • Extended 9,720-run sweep ($K \in \{32, 48, 64\}$, ≈ 4 min): pythonexperiments/big_
- 681 experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK
- 682 • Q/R/M/C stage analysis and bootstrap CIs: pythonexperiments/big_experiment/
- 683 analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_
- 684 dirtmp/big_experiment_qrmc_40
- 685 • Multi-agent oracle interventions: pythonexperiments/big_experiment/exp_oracle_
- 686 interventions.py--workers8--out_dirtmp/big_experiment_oracle
- 687 • Phase 1–4 smoke tests: pythonscripts/run_all_smokes.py--out_dirtmp/all_smokes

688 I Asset and license note

689 The experiments use MiniGrid and BabyAI, both released under the MIT License. The locally
690 implemented baselines and benchmark runner scripts introduced in this paper will be released under
691 the MIT License as part of the camera-ready artifact package.

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